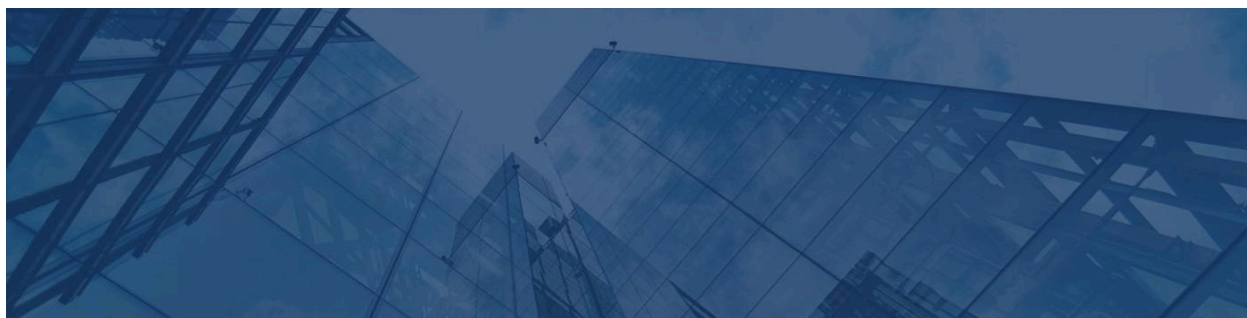




Lead AI Engineer

Artificial Intelligence, Machine Learning & Intelligent Platform Leadership

-  Remote / Hybrid / Onsite
 -  Full-Time or Part-Time
 -  The Copia Group, LLC
 -  Lead Level
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About The Copia Group

The Copia Group is an investment management firm supporting innovative companies across AI, fintech, blockchain, digital infrastructure, and emerging technology sectors.

Through strategic investments and partnerships, we help accelerate the growth of technology-driven businesses building intelligent applications, autonomous systems, enterprise AI platforms, and next-generation software solutions. Our portfolio companies leverage machine learning, generative AI, and advanced analytics to solve complex business challenges and create transformative customer experiences.

We are seeking an experienced **Lead AI Engineer** to lead the design, development, and deployment of cutting-edge AI solutions while mentoring engineering teams and driving AI strategy across multiple products.

The Opportunity

As a Lead AI Engineer, you will combine hands-on AI development with technical leadership to build production-ready machine learning systems and generative AI applications. You'll collaborate closely with Product Managers, Data Scientists, Software Engineers, DevOps, and Executive Leadership to deliver scalable AI-powered products that drive innovation and business growth.

This role offers significant ownership over AI architecture, MLOps strategy, and engineering best practices while providing opportunities to mentor engineers and shape the organization's long-term AI roadmap.

What You'll Do

AI & Machine Learning Development

- Design, develop, and deploy production-grade machine learning models and AI applications
- Build intelligent systems using Large Language Models (LLMs), deep learning, and predictive analytics
- Develop scalable AI APIs, inference services, and model-serving infrastructure
- Optimize AI model performance, accuracy, latency, and operational efficiency
- Build reusable AI components that accelerate product development

Generative AI & LLM Engineering

- Design and implement applications powered by Large Language Models
- Develop Retrieval-Augmented Generation (RAG) systems and AI knowledge platforms
- Build AI agents, copilots, workflow automation, and conversational AI solutions
- Fine-tune foundation models for domain-specific business use cases
- Evaluate emerging AI models and recommend adoption strategies

Technical Leadership

- Define AI architecture, engineering standards, and best practices
- Mentor AI Engineers, Machine Learning Engineers, and Software Engineers
- Conduct technical design reviews and provide architecture guidance
- Drive experimentation, research, and innovation across AI initiatives
- Lead technical planning and AI roadmap execution

MLOps & Cloud Engineering

- Design scalable MLOps pipelines for training, deployment, monitoring, and model lifecycle management
- Build cloud-native AI infrastructure using containers and Kubernetes
- Implement CI/CD pipelines for AI applications and model deployment
- Monitor production AI systems for reliability, drift, and performance
- Ensure security, governance, and responsible AI practices across all solutions

Cross-Functional Collaboration

- Partner with Product Managers to translate business goals into AI-powered products
 - Work closely with Data Engineers to build scalable data pipelines and feature stores
 - Collaborate with DevOps teams to optimize cloud infrastructure and deployment automation
 - Communicate AI strategy and technical direction to leadership and stakeholders
 - Foster a collaborative culture focused on innovation, experimentation, and continuous learning
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What We're Looking For

Required Qualifications

Experience

- 7+ years of professional software engineering or machine learning engineering experience
- 4+ years of hands-on AI/ML development in production environments
- 2+ years of technical leadership or lead engineering experience
- Proven experience delivering enterprise AI applications at scale
- Experience working within Agile software development environments

Technical Expertise

Strong experience with:

- Python
- PyTorch or TensorFlow
- Scikit-learn
- Hugging Face Transformers
- FastAPI or Flask

- SQL and data processing frameworks
- Docker
- Kubernetes
- Git

Strong experience with AI technologies:

- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- AI Agents
- LangChain or LlamaIndex
- Vector Databases
- Prompt Engineering
- Model Fine-Tuning
- Embedding Models

Strong understanding of:

- Deep Learning
- Natural Language Processing (NLP)
- MLOps and Model Lifecycle Management
- Distributed Systems
- Cloud-Native AI Infrastructure
- Responsible AI and Model Governance
- API Design and Microservices
- CI/CD and DevOps Practices

Preferred Qualifications

- Experience developing enterprise AI copilots, intelligent assistants, or autonomous agent systems
- Experience with OpenAI, Anthropic Claude, Google Gemini, Cohere, or open-source LLMs
- Familiarity with vector databases such as Pinecone, Weaviate, Milvus, Chroma, or pgvector
- Experience deploying AI workloads on AWS, Azure, or Google Cloud Platform
- Knowledge of MLflow, Kubeflow, Airflow, SageMaker, or Vertex AI
- Experience working in fintech, healthcare, SaaS, cybersecurity, or blockchain industries
- Contributions to open-source AI projects, published research, or technical conferences

Technology Environment

AI & Machine Learning

Python • PyTorch • TensorFlow • Scikit-learn • Hugging Face • ONNX

Generative AI

OpenAI • Anthropic Claude • Google Gemini • LangChain • LlamaIndex • AI Agents • RAG

Data & Storage

Pandas • NumPy • PostgreSQL • Redis • Pinecone • Weaviate • Chroma • pgvector

Cloud & Infrastructure

AWS • Azure • Google Cloud Platform • Docker • Kubernetes • MLflow • Terraform

DevOps & Monitoring

GitHub Actions • Airflow • Kubeflow • Prometheus • Grafana • OpenTelemetry

Tools & Collaboration

GitHub • Jira • Slack • Notion • Figma

Why Join Us

- ✦ Flexible remote, hybrid, and onsite work options
- ✦ Full-time and part-time opportunities
- ✦ Competitive compensation package
- ✦ Opportunity to lead the development of next-generation AI platforms and intelligent products
- ✦ Exposure to cutting-edge AI, blockchain, fintech, cloud-native technologies, and emerging innovations
- ✦ Collaborative, engineering-first culture focused on innovation, ownership, and technical excellence
- ✦ Career advancement into Principal AI Engineer, AI Architect, Engineering Manager, Director of AI Engineering, or Chief AI Officer

✨ Strong investment in leadership development, AI research, continuous learning, conference participation, and open-source contributions

Apply

If you are passionate about building production-grade AI systems, leading high-performing engineering teams, and creating intelligent solutions that solve real-world problems, we'd love to connect with you.